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CSC_360
Class Notes

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- Introduction to Computer Graphics and Digital Image Processing

- Difference between Computer Graphics and Image Processing:
  1) Computer Graphics deals with Creation & Rendering of images via algorithms & programming.
  2) Image Processing deals with Analysis & Modification of digital images.

- Geometric Primitives:
  The most basic elements used to draw graphical pictures include:
  ~ Lines
  ~ Curves
  ~ Areas
  Complicated graphics are created by combining such primitives.

- Java Graphics Framework:
  Java offers several frameworks to develop graphical applications:
  ~ Java Swing – desktop GUI applications.
  ~ JavaFX – modern framework for user interfaces, animations, charts & multimedia.
  ~ Java 2D – API for drawing shapes, text, colors, images & customized graphics.

- Event-Driven Programming:
  ~ User interface programming follows the Event-Driven Programming concept.
  ~ Any user activities (button press, mouse motion, keyboard typing, etc.) result in Events generation.
  ~ Such events are detected by Event Listeners, which trigger corresponding actions.

- Static & Interactive Graphics:
  ~ Static Graphics: images that are not altered after being drawn.
  ~ Interactive Graphics: images that react to user actions.

- Curves and Calculus:
  ~ Representation through mathematical equations.

~ Used to determine slope, tangents, curvature, and smoothness of connection between two points.
~ It is the mathematical basis for curve algorithms.

- Raster Graphics VS Vector Graphics:

  1) Raster Graphics
  ~ It is pixel based.
  ~ Resolution Dependent.
  ~ Best used for photography.

  2) Vector Graphics
  ~ Created using mathematical equations.
  ~ Resolution Independent.
  ~ Best suited for logos, icons, and illustrations.

- Protocols to Access GitHub Repository:

  1) HTTPS Protocol
  ~ Uses username and personal access token for authentication.
  ~ Authentication required every time you access repositories.

  2) SSH Protocol (Secure Shell)
  ~ Uses a pair of cryptographic keys for authentication.
  ~ It is more secure and easy after initial set up.
  ~ No need to authenticate every time.

- SSH Key Creation:

  Commands: **ssh-keygen**
  ~ The keys are generated in the .ssh folder in the user's home directory.
  ~ To navigate to the home directory use: cd ~

- Files Generated:
  ~ Public Key (.pub file): Given to GitHub server for authentication.
  ~ Private Key: Should be kept secure in the local system and never be shared anywhere else.

- Tools Used:
  ~ Git Bash
  ~ ssh
  ~ ssh-keygen
  ~ GitHub